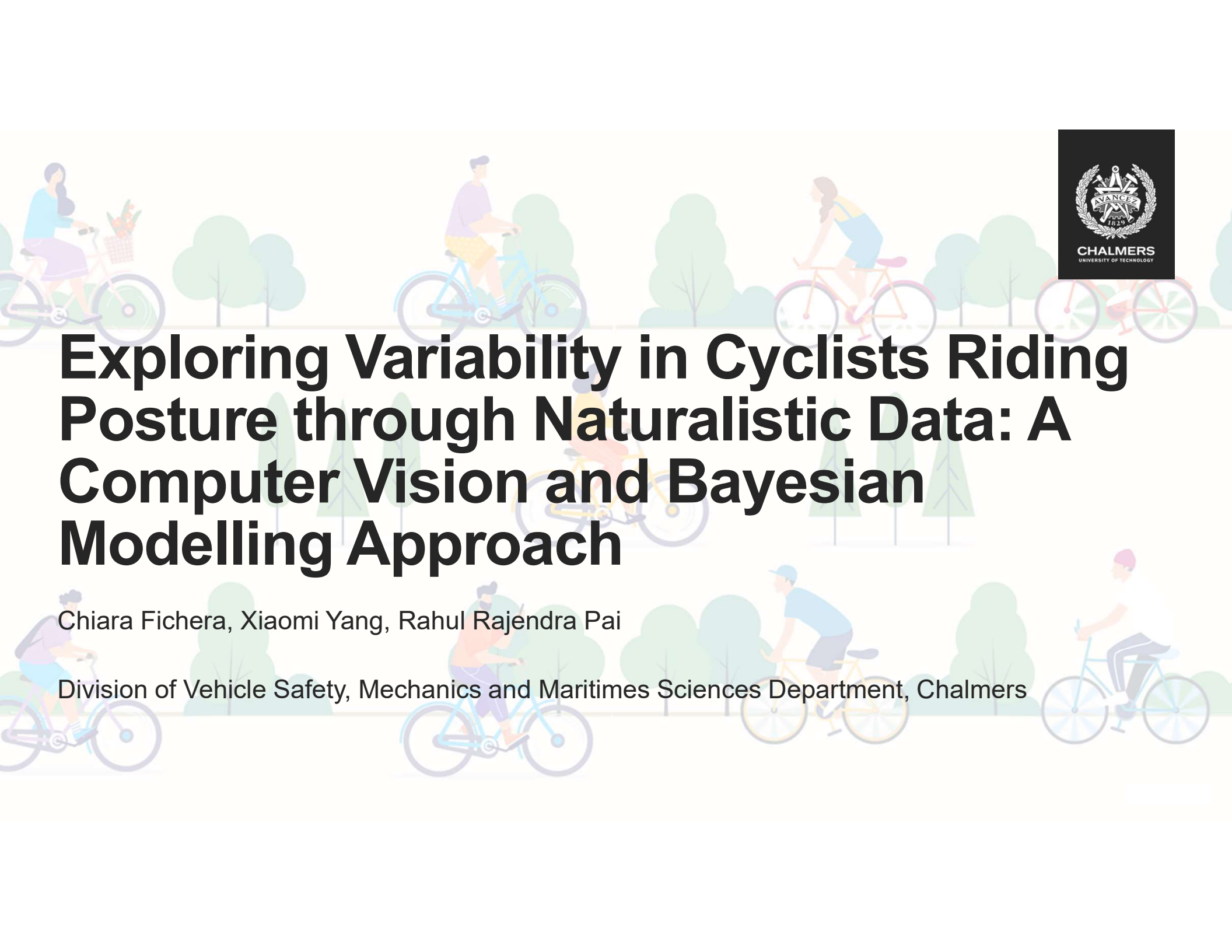




Exploring Variability in Cyclists Riding Posture through Naturalistic Data: A Computer Vision and Bayesian Modelling Approach

Chiara Fichera, Xiaomi Yang, Rahul Rajendra Pai

Division of Vehicle Safety, Mechanics and Maritimes Sciences Department, Chalmers

Introduction



The severity and type of injuries sustained by cyclists can be influenced by various factors such as riding posture, helmet usage, bike type, and speed.

By understanding how people interact with the vehicle, we can understand initial boundary conditions during a crash and are able to design prevention systems.

Introduction

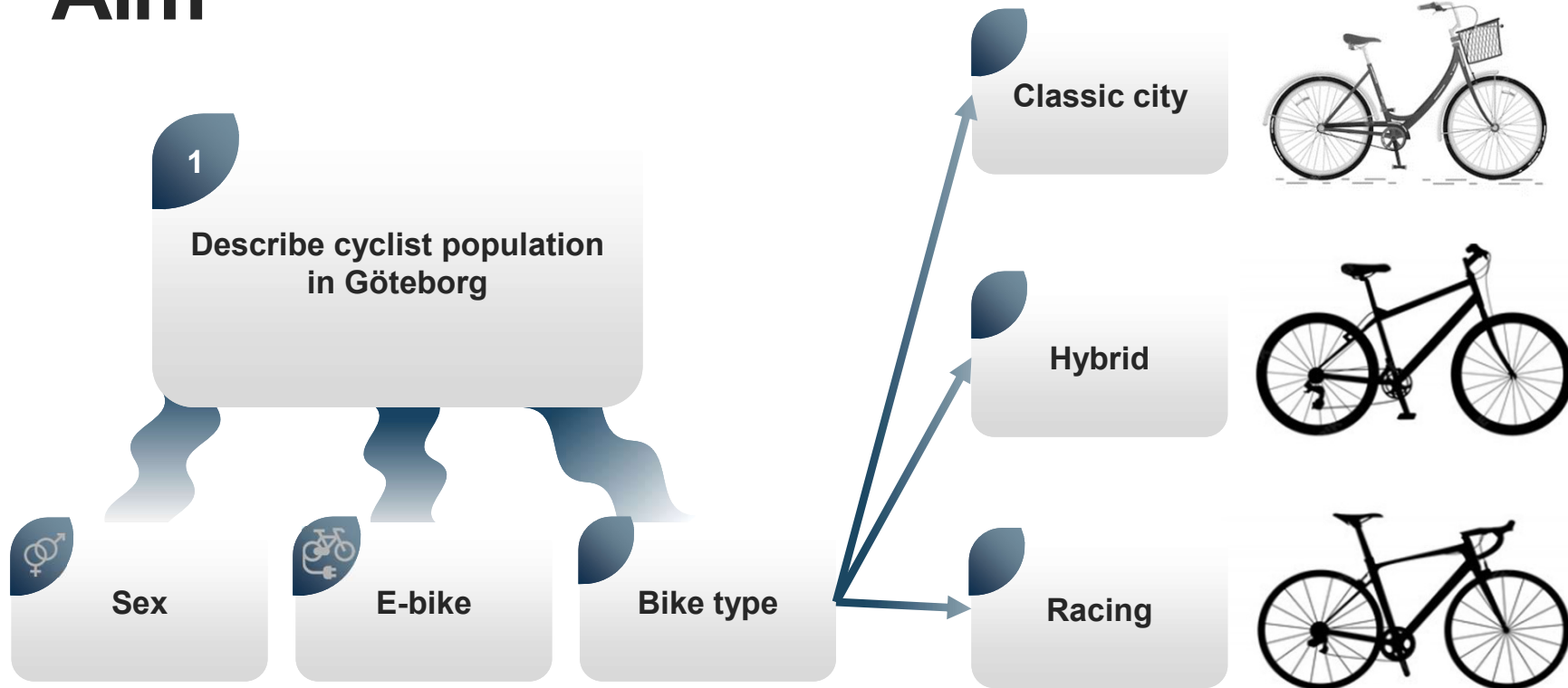


The severity and type of injuries sustained by cyclists can be influenced by various factors such as **riding posture**, **helmet usage**, **bike type**, and speed.

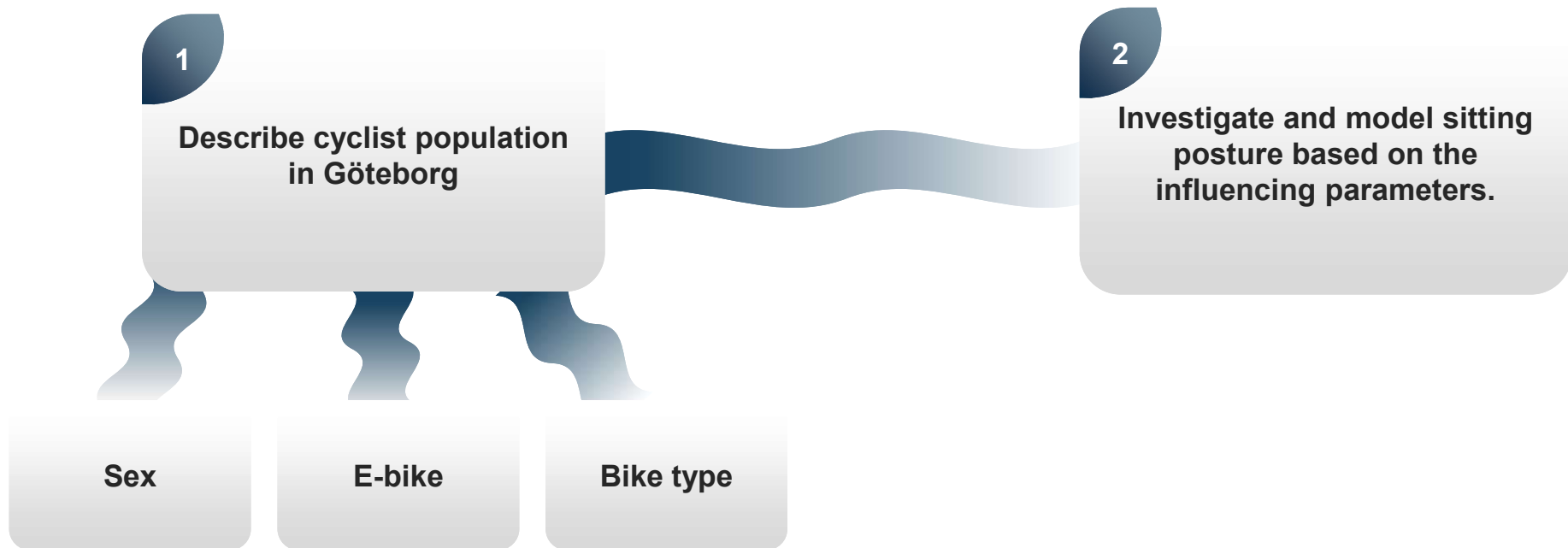
By understanding how people interact with the vehicle, we can understand initial boundary conditions during a crash and are able to design prevention systems.



Aim



Aim



Methods



Collect and extract data by video tracking algorithms and annotations.

Descriptive statistics to analyze variability of cyclists

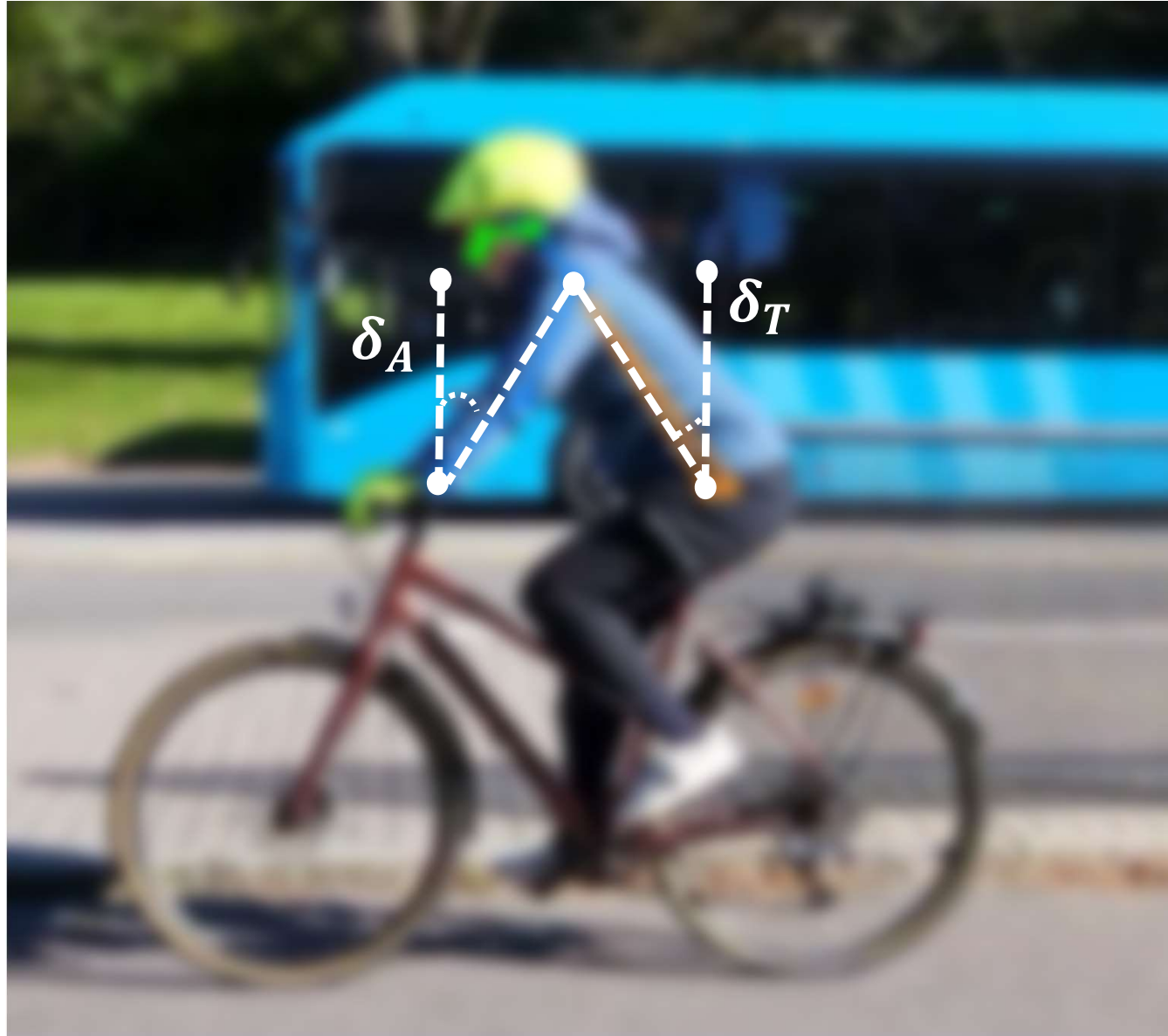
Develop Bayesian regression to analyze and model cyclist posture

Predict sitting position of cyclist based on the developed model

Data Collection

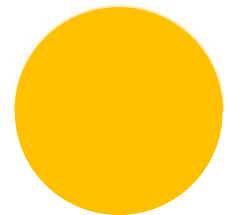
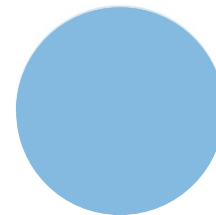
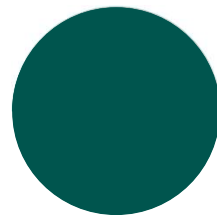
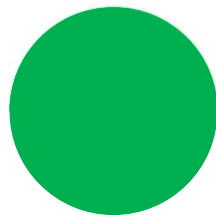
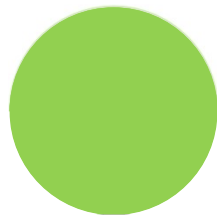
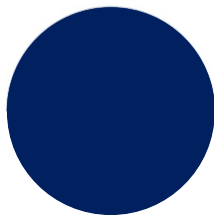
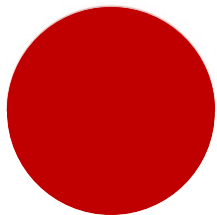


**Compute
Torso
&
Arm angles**



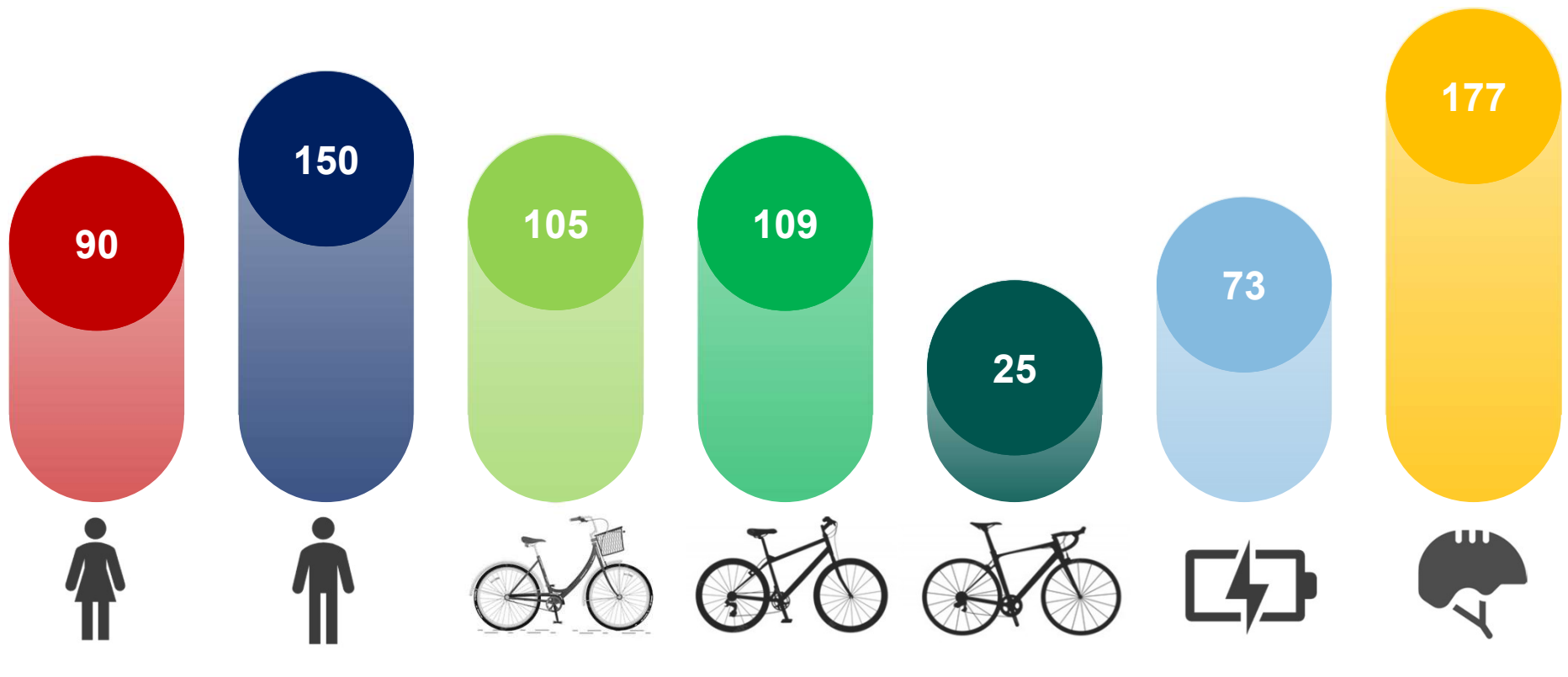
Results

Total sample: 240 Cyclists

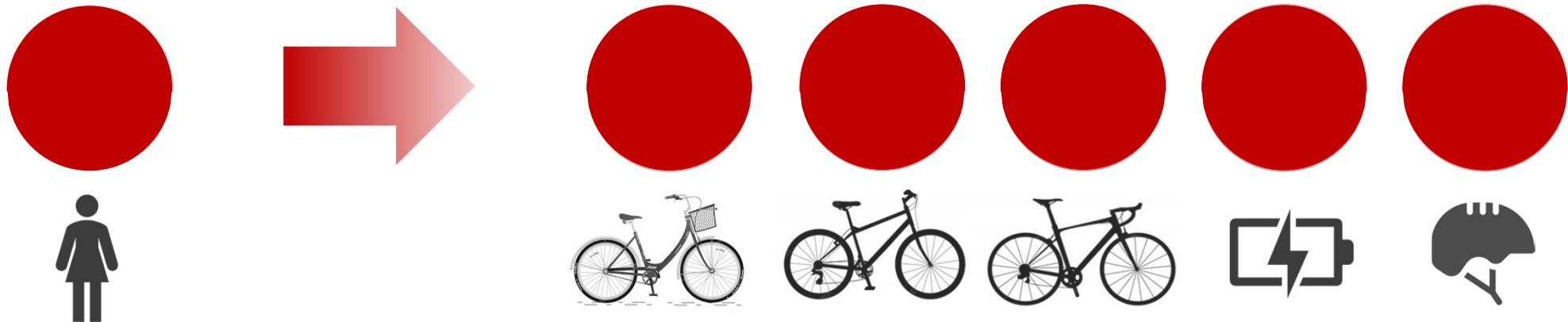


Results

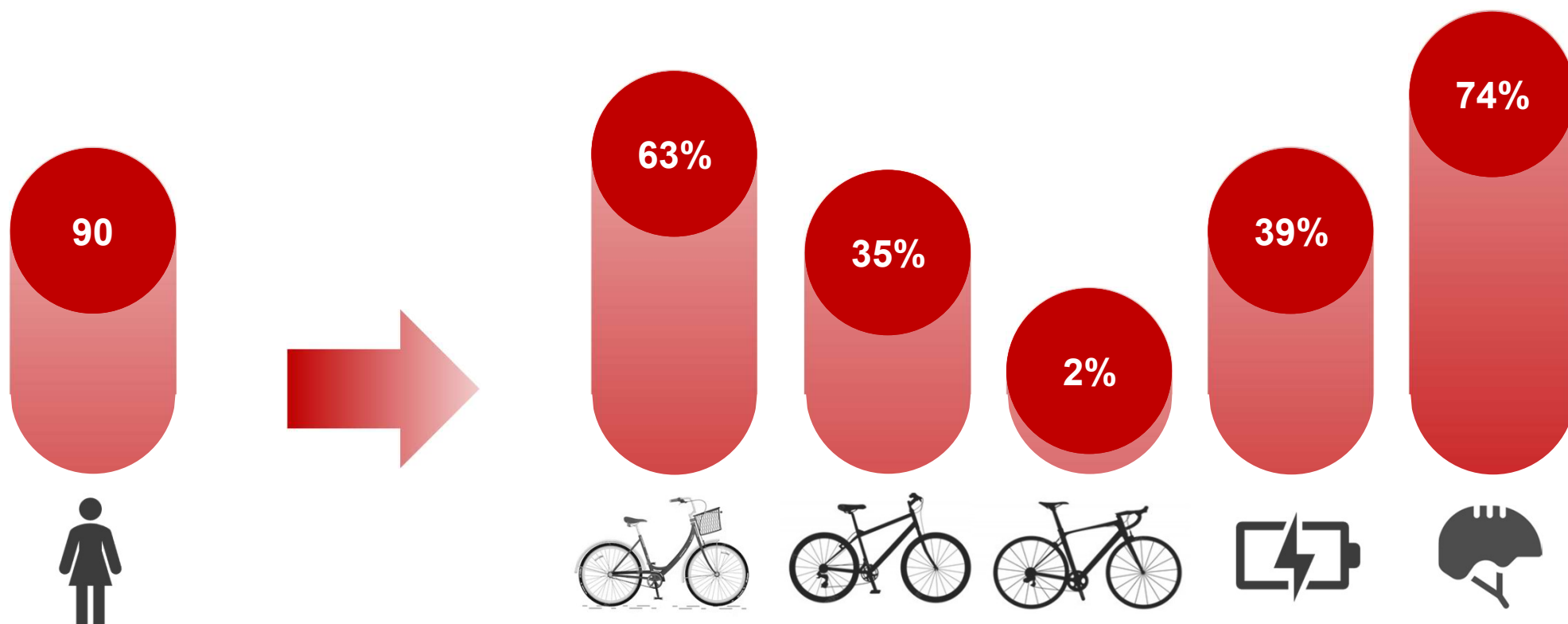
Total sample: 240 Cyclists



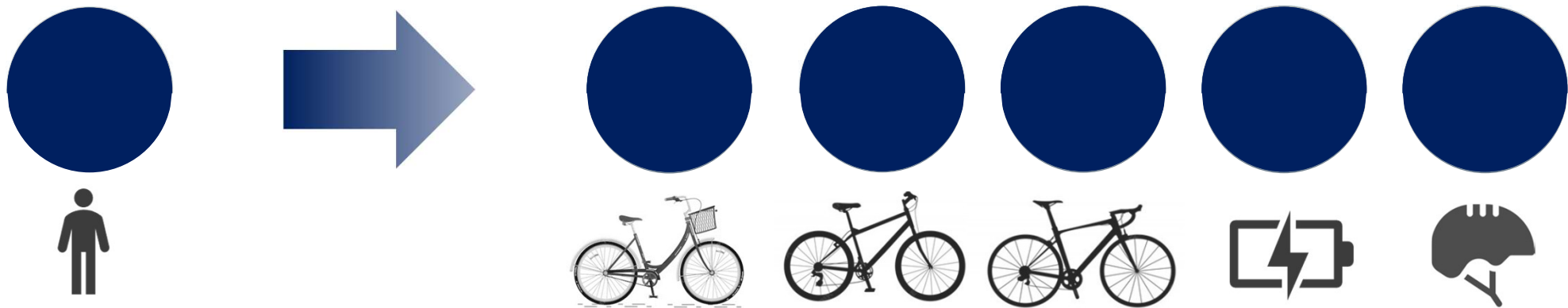
Females vs. Bike type, E-bike, Helmet



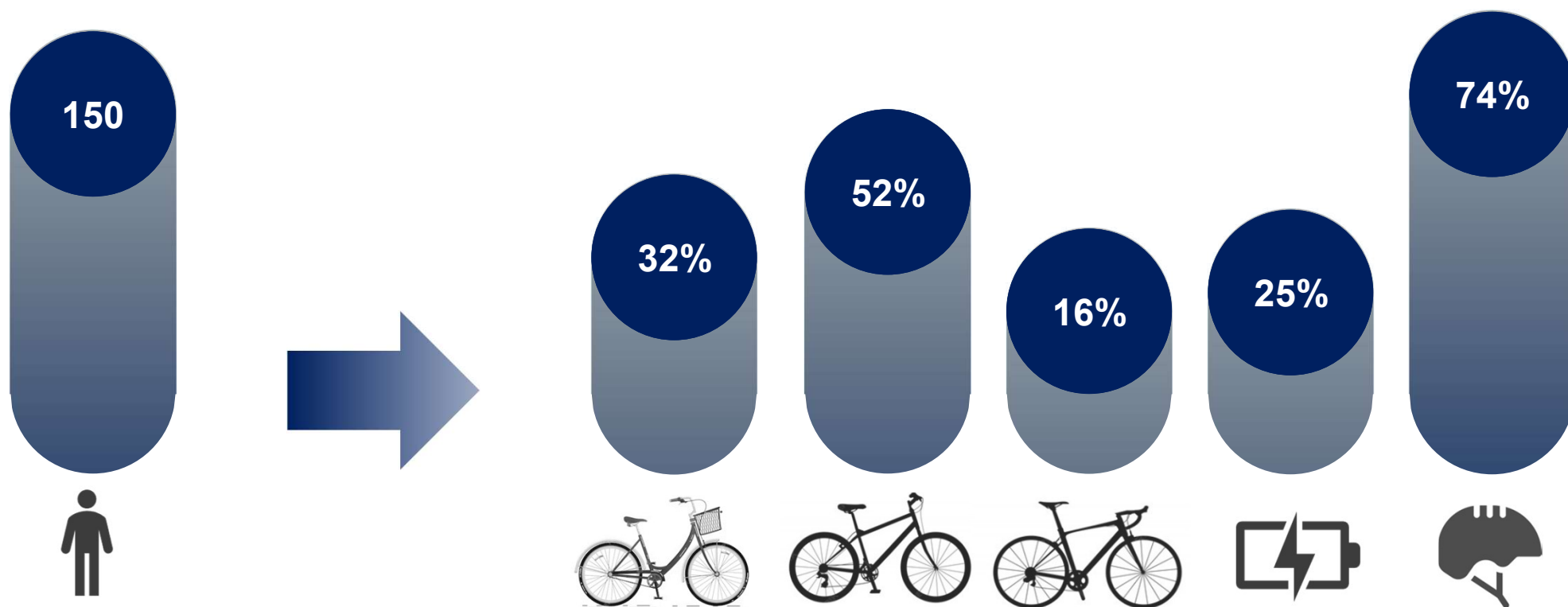
Females vs. Bike type, E-bike, Helmet



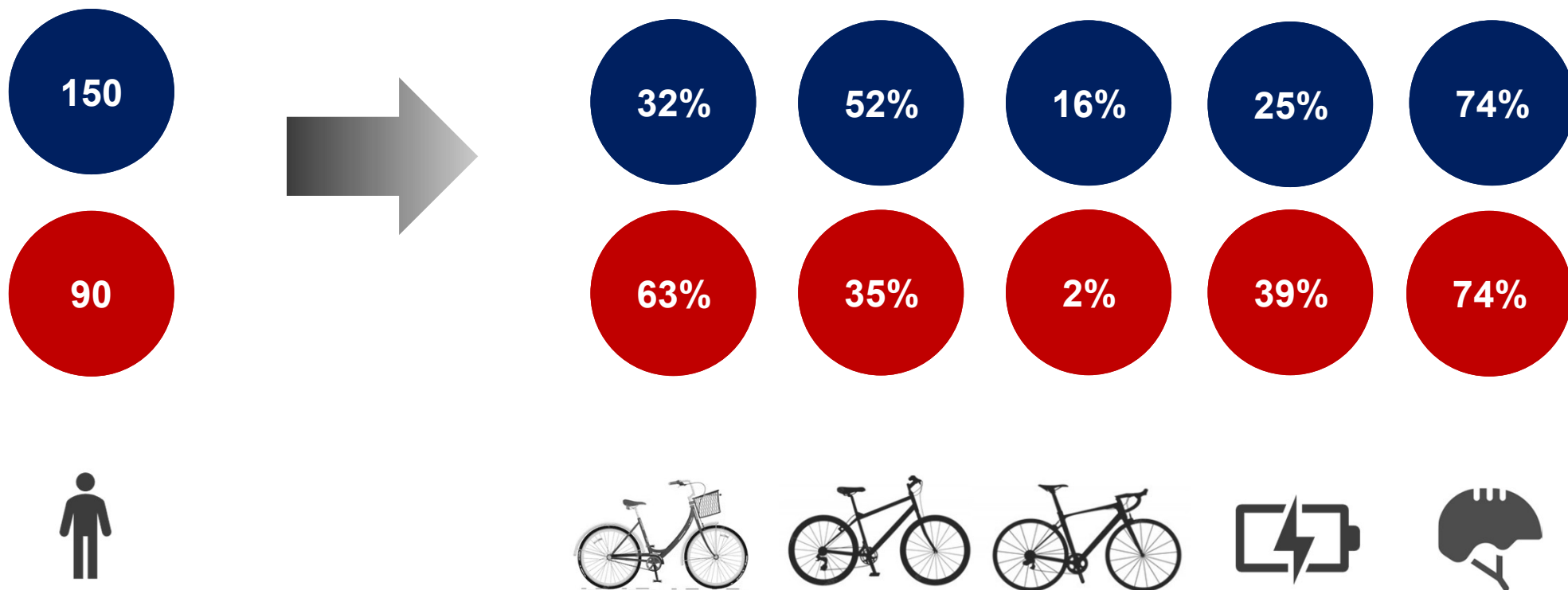
Males vs. Bike type, E-bike, Helmet



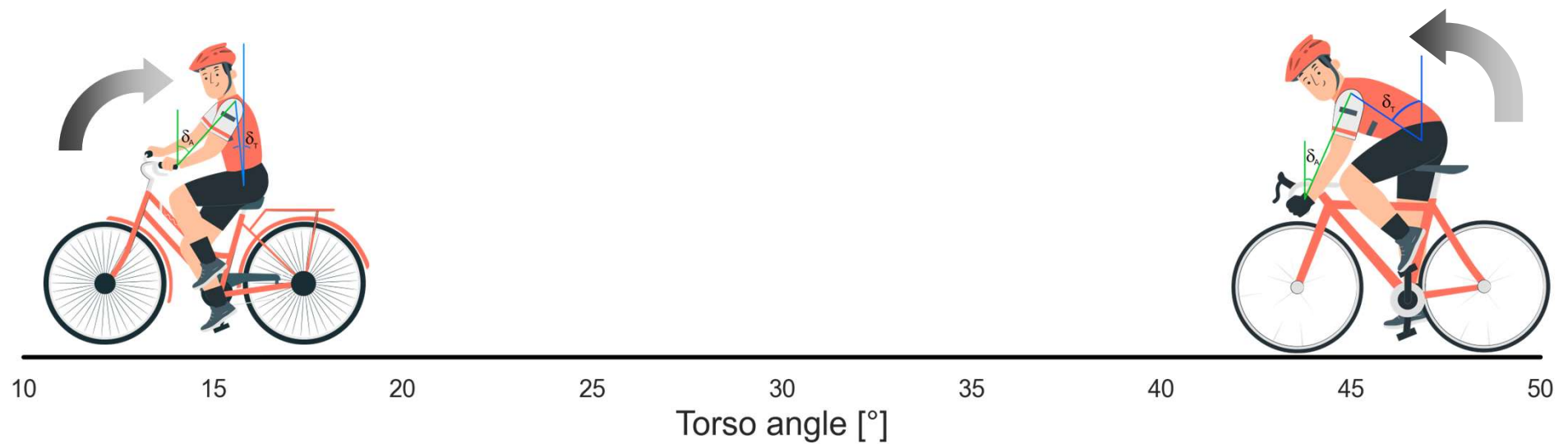
Males vs. Bike type, E-bike, Helmet



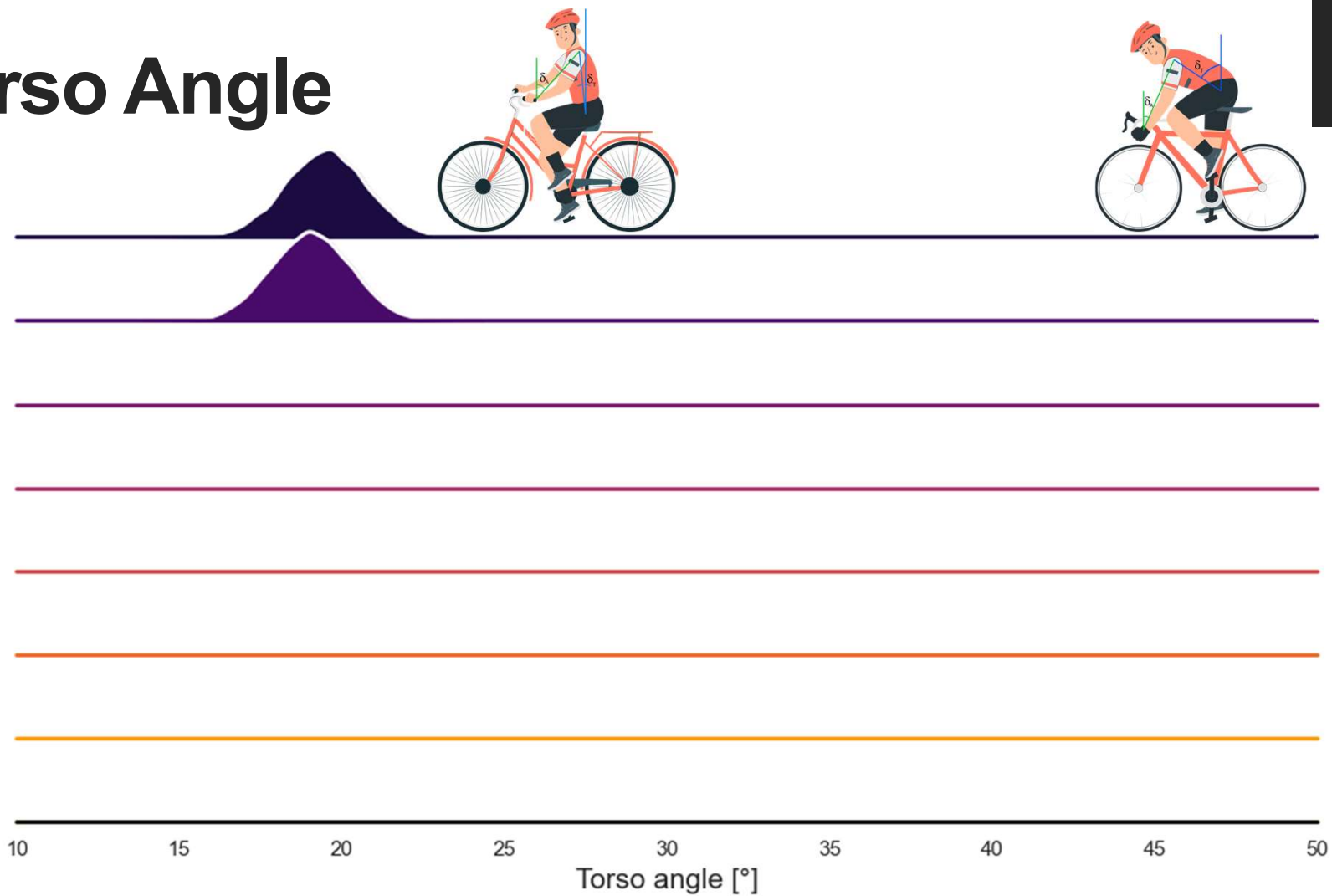
Comparison



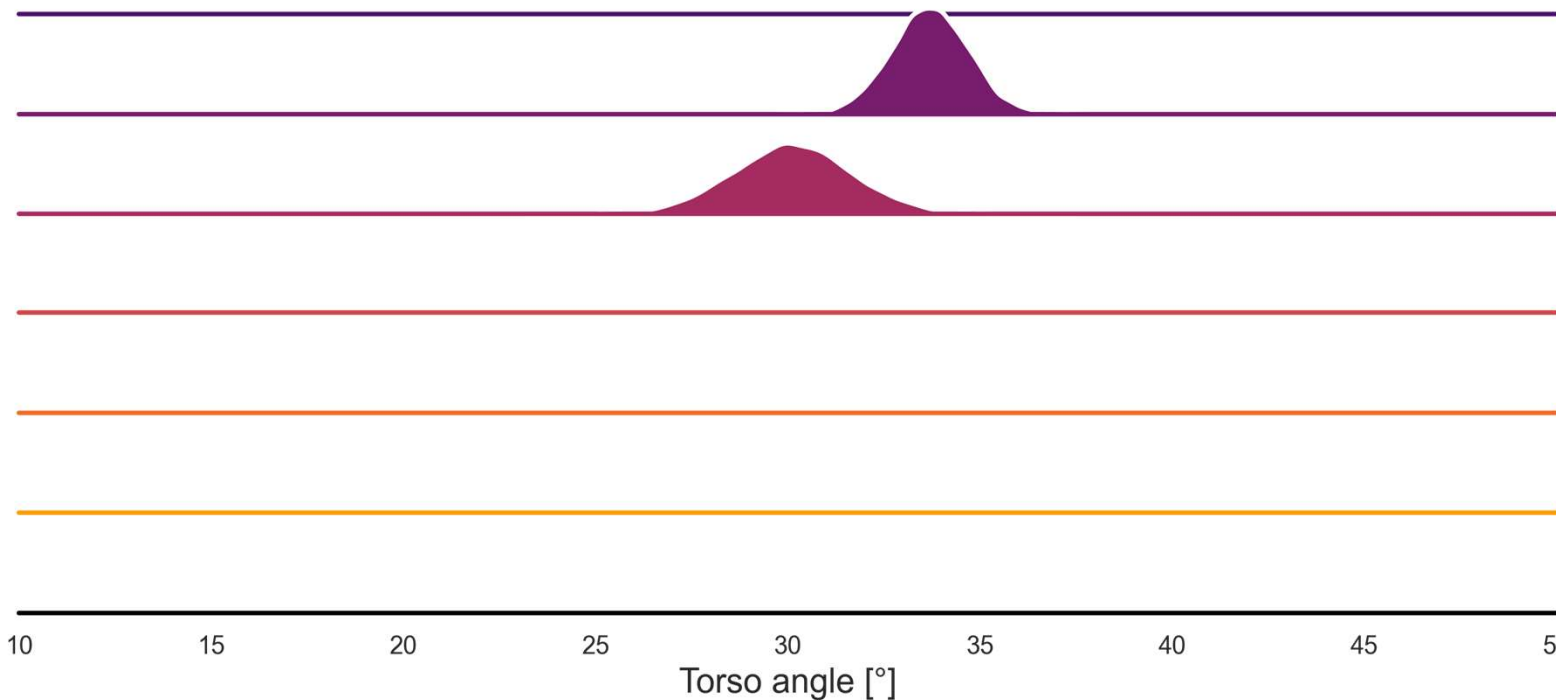
Torso Angle



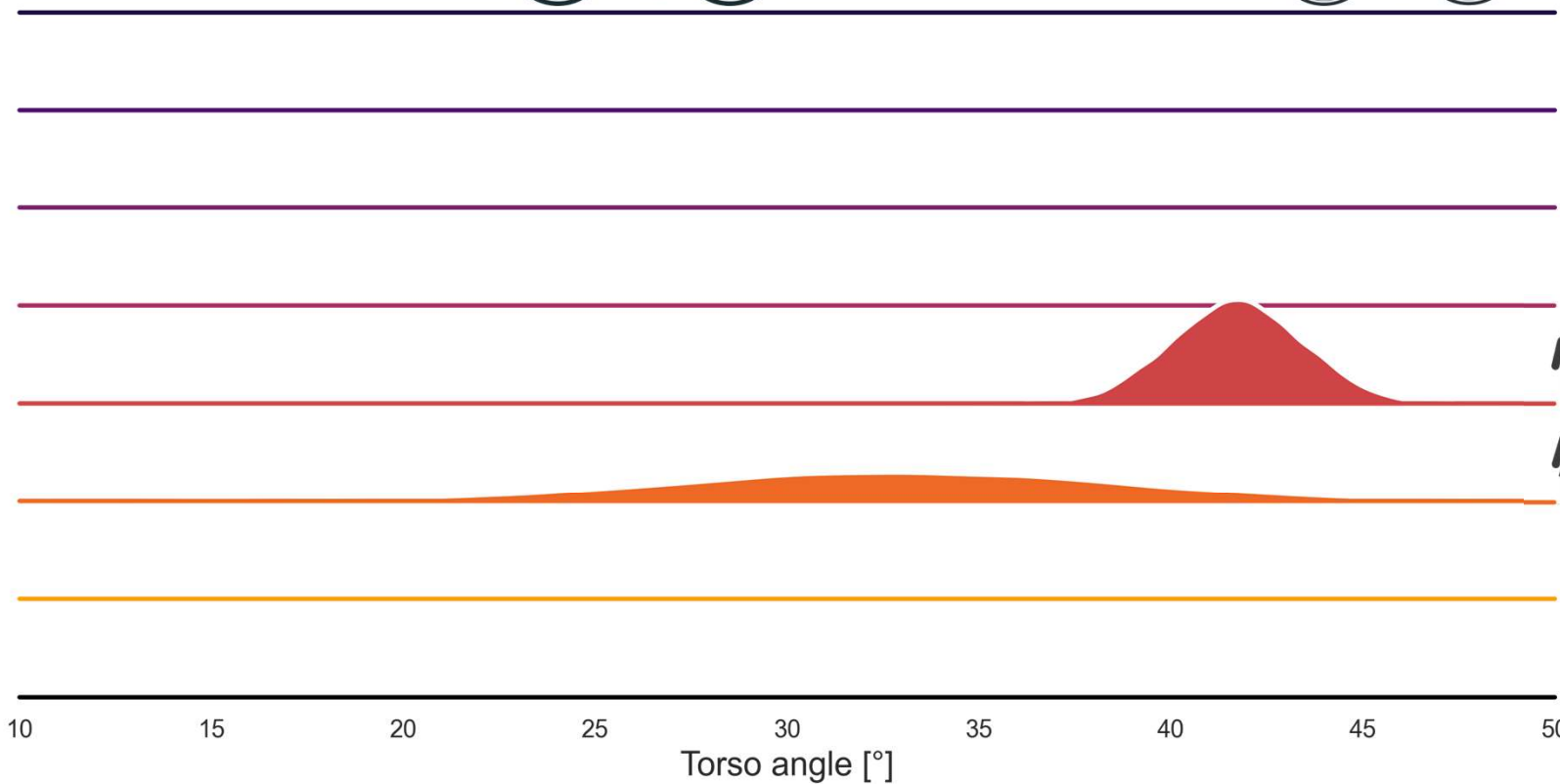
Torso Angle



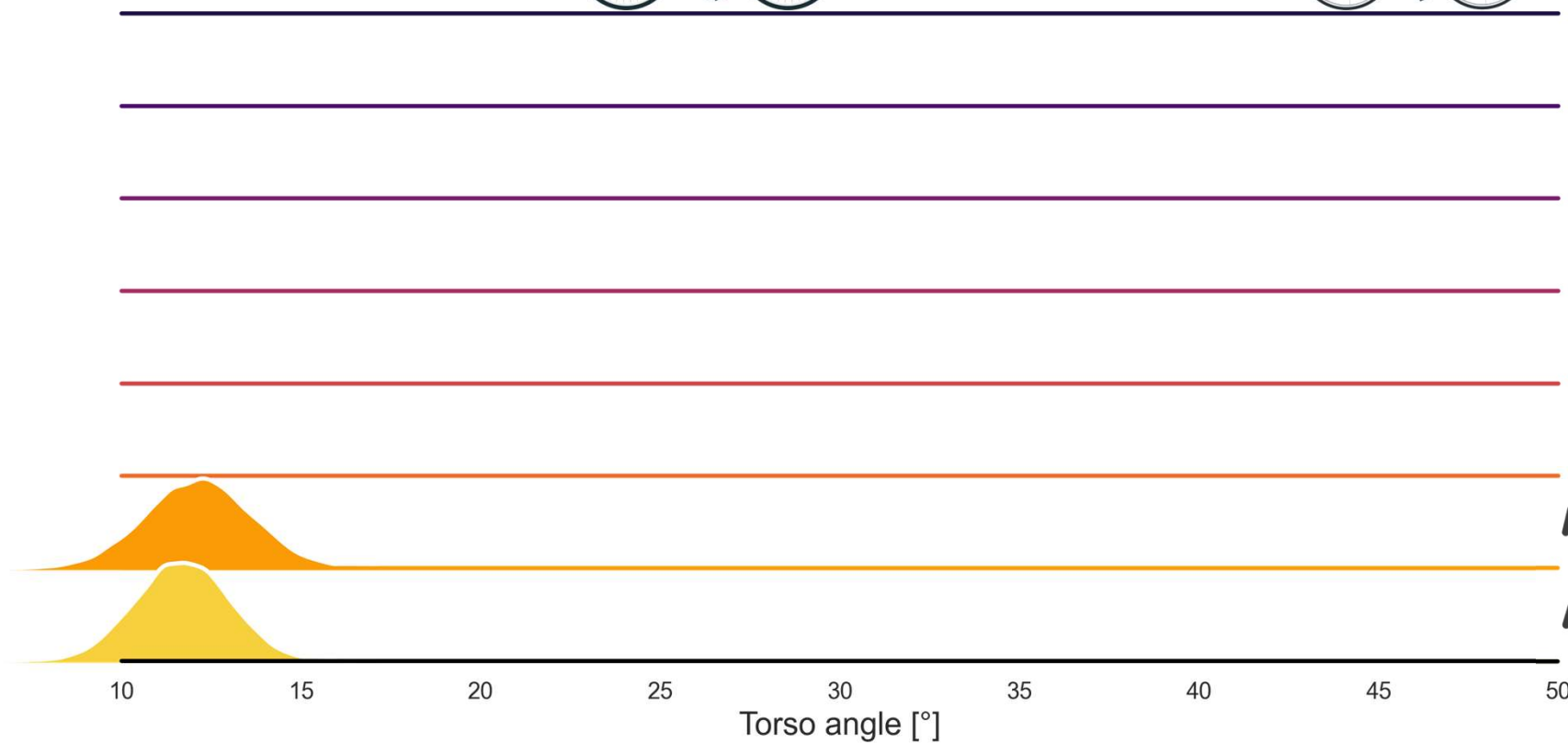
Torso Angle



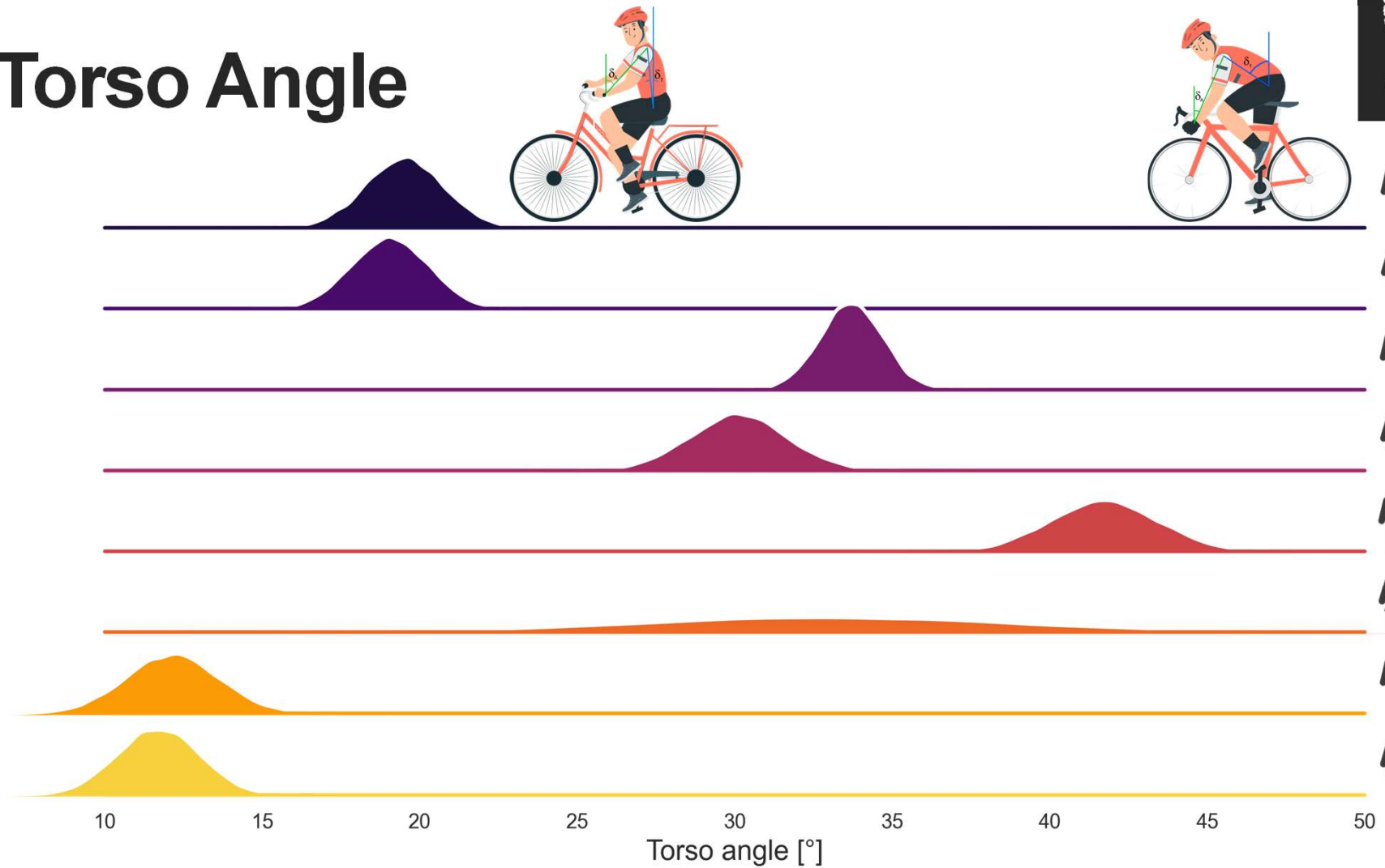
Torso Angle



Torso Angle

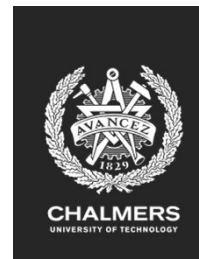
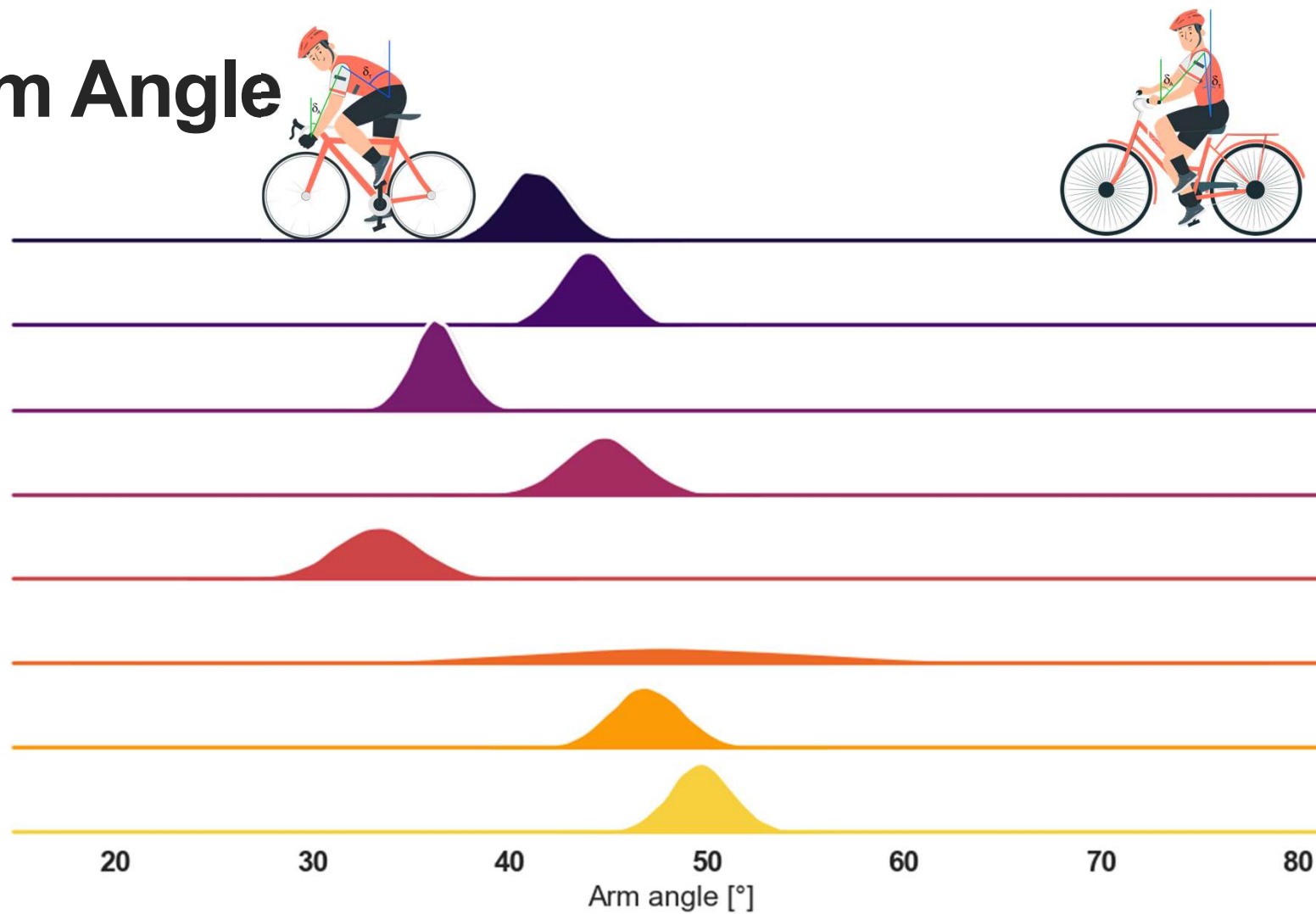


Torso Angle



2024-10-28

Arm Angle



Summary



150 of 240 were males.



Most common used bikes are classic or hybrid.



30% are e-bikes.



75% wear the helmet.



More females ride a classic or e-bike



Males ride more hybrid/racing bikes.

Summary

- Quantify the angles for each bike type.
- Sex differences are distinct for racing and hybrid configurations.
- Being on an ebike, resulted in the most upright torso position.
- Arm angle varies between 40-50% for all the configurations.





CHALMERS
UNIVERSITY OF TECHNOLOGY